



SME Risk Intelligence Platform

AI-powered credit risk scoring & explainability for small & medium enterprise lending

TRACK

AI / Data Science Track

JOB PROFILE

AI Engineer / Data Scientist

DEPI · Digital Egypt Pioneers Initiative

PROBLEM STATEMENT

Banks and microfinance lenders assess SME creditworthiness manually — a process that is slow, inconsistent across analysts, and hard to justify to applicants or regulators, leaving high-potential businesses under-served.

THE IDEA

A machine-learning risk engine that scores any SME loan application in seconds from its financial indicators, explains *why* using SHAP, and delivers a bilingual, actionable report — turning credit risk into a transparent, repeatable decision.



ROC-AUC · MODEL
CONFIDENCE

Accuracy	92.11%
F1-Score	91.48%
Precision	92.97%
Recall	90.03%

METHODOLOGY

- 01

Data Engineering
35,000 SME records cleaned, winsorized & expanded to 12 engineered risk features
- 02

Model Benchmarking
7 classifiers benchmarked with 10-fold stratified cross-validation on ROC-AUC
- 03

Hyperparameter Tuning
RandomizedSearchCV optimizes each candidate before final selection
- 04

Calibration & Explainability
Isotonic probability calibration + SHAP TreeExplainer per prediction
- 05

MLOps & Deployment
MLflow logs every run; best model auto-promoted to Production

OUTPUTS & DELIVERABLES

- Instant risk score with SHAP-based explanation
- Bilingual (AR/EN) PDF risk report, RTL-ready
- Gemini AI strategic lending advisory
- Batch CSV scoring for full loan portfolios
- Live analytics dashboard & model comparison

STAKEHOLDERS

- Banks & Lenders
- Microfinance Institutions
- Credit Analysts
- SME Owners
- Regulators

MODELING & ML

- 7-model comparative benchmark
- RandomizedSearchCV tuning
- Isotonic probability calibration
- Feature Importance Selection
- 10-fold stratified CV

TRUST & EXPLAINABILITY

- SHAP TreeExplainer per case
- MLflow experiment tracking
- Model Registry (auto-promotion)
- Confidence & margin scoring
- Global vs. Local Interpretability

OUTPUTS & FEATURES

- Bilingual Arabic/English + RTL
- Gemini AI advisory engine
- ReportLab PDF generation
- Batch & single-record scoring
- Automated Executive Summary
- What-If Scenario Simulation

TOOLS & DEPLOYMENT

- Python & Pandas
- Scikit-learn
- Supabase Database ACTIVE
- Streamlit Application LIVE
- Streamlit Cloud Deployment DEPLOYED
- Next Phase: FastAPI + React IN PROGRESS
- SHAP Explainability
- MLflow Tracking